

Erythropoietin (EPO)

Erythropoietin is a type of protein called a growth factor. It is used to treat a low number of red blood cells (anaemia) due to cancer or its treatment.

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What is erythropoietin?

Erythropoietin is a type of protein called a growth factor. It stimulates the [bone marrow](#) to make red blood cells. Red blood cells contain a protein called haemoglobin (Hb). It carries oxygen around the body.

The body makes erythropoietin naturally. But it is also made as a drug to treat a low number of red blood cells (anaemia).

Anaemia can cause symptoms such as:

- pale skin
- lack of energy
- feeling breathless
- feeling dizzy and light-headed.

Anaemia can be caused by cancer, or by cancer treatments such as [chemotherapy](#).

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

When is erythropoietin used?

Doctors usually treat anaemia with a blood transfusion to replace red blood cells. But they may use erythropoietin instead if:

- the cancer treatment means blood transfusions are needed often
- blood transfusions are not suitable
- the anaemia is caused by a type of blood cancer called [myelodysplasia](#).

There are different types of erythropoietin treatments. They are often called EPO. Treatments include:

- epoetin alfa – for example, Eprex®
- epoetin beta – for example, NeoRecormon®
- epoetin zeta – for example, Retacrit®
- darbepoetin alfa (Aranesp®).

Your doctor will talk to you about erythropoietin and its possible side effects before you agree ([consent](#)) to have treatment.

How erythropoietin is given

Erythropoietin is usually given as an injection under the skin (subcutaneously). It is usually given in your thigh or tummy area. You, or someone who looks after you, may be shown how to give the injections. Or they can be given by a nurse that sees you at home or at the GP surgery.

Storing erythropoietin

Erythropoietin should be stored in the fridge in its original packaging. Take the injection out of the fridge 30 minutes before you have it, so it can come up to room temperature.

Always follow the storage advice given by your pharmacist.

Your course of treatment

If you have anaemia caused by chemotherapy, you may have erythropeotin 3 times per week. But this can vary depending on why you are having it and what has been prescribed.

You will have regular blood tests to check how you are responding to treatment. These tests will measure your level of red blood cells and haemoglobin (Hb). Your doctor may change the dose of erythropoietin depending on these results.

If blood tests show the treatment is working, you usually continue having injections until 1 month after you finish your course of chemotherapy.

During treatment, your doctor may also give you iron tablets or injections. Your body needs iron to make red blood cells. Your doctor can tell you if this will be helpful for you.

Your nurse, pharmacist or doctor will discuss your treatment plan with you.

About side effects

When you have erythropoietin injections, the amount of the protein in your body becomes much higher than normal. This may cause side effects, but these are usually mild or moderate.

We explain the most common side effects of this treatment here. We also include some less common side effects.

You may get some of the side effects we mention, but you are unlikely to get all of them. If you are also having treatment with other cancer drugs, you may have some side effects that we have not listed here. If you have chemotherapy with erythropoietin, some side effects may be worse. We have more information about [chemotherapy](#). Always tell your doctor, nurse or pharmacist about any side effects you have.

Your doctor can give you drugs to help control some side effects. It is important to take them exactly as your nurse or pharmacist explains. This means they will be more likely to work for you. Your nurse will give you advice about managing your side effects. After your treatment is over, most side effects start to improve.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects when treatment is being given

Skin irritation at the injection site

You may get side effects in the area that the injections are given. Having the injection in a different place each time can help with this.

The skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin.

If this happens, let your doctor know. Painkillers may help.

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment. Because there are different types of EPO the frequency of side effects may vary a little.

Feeling sick

Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. [If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
-

drinking enough fluids to keep you hydrated and to replace lost salts and minerals

- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Flu-like symptoms

Some people have flu-like symptoms, such as:

- having a fever (high temperature)
- chills
- feeling sick
- muscle and joint pains
- weakness
- dizziness
- tiredness.

You are more likely to have these symptoms at the start of your treatment. Contact the hospital on the number you have been given if you have any of these symptoms. Your doctor, nurse or pharmacist can give you advice.

They can suggest treatments to help.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Cough

This treatment may cause a cough. Tell your doctor or nurse if you notice this. They may suggest treatments to help.

Skin rash

You may develop a rash, which may be itchy and red. Tell your doctor if you notice this. They can give you advice and may prescribe creams or medicines to help.

Serious skin changes

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

High blood pressure

Erythropoietin may cause high blood pressure. Tell your doctor or nurse if you have ever had any problems with your blood pressure. You should also tell them if you take any medication for high blood pressure. Your nurse will check your blood pressure regularly during your treatment.

Signs that your blood pressure may be very high include:

- a severe headache
- feel drowsy or confused
- problems with your eyesight
- fits (seizures)
- being sick (vomiting)

If you have any of these symptoms, contact the hospital straight away on the 24-hour contact number you have been given. If you cannot get through to your doctor, call the **NHS urgent advice** number on **111**.

Increased risk of stroke

Erythropoietin may increase the risk of a mini-stroke or stroke, but this is not common. A stroke is when blood stops flowing to a part of your brain. This can affect things like speech and movement. Call 999 or go to the hospital straight away if you or someone else notices:

- you are confused
- you have difficulty speaking
- your face drooping
- numbness or weakness on one side of your body.

Other important information

Because this treatment is a growth factor, it is thought that it may cause cancer to progress in some people. Your doctor will consider the benefits and risks of this treatment before deciding whether it is suitable for you. Your doctor will talk to you about this. If you have any concerns or questions, always talk to your doctor or nurse.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Contraception

Your doctor, nurse or pharmacist may advise you not to get pregnant or make someone pregnant while having this treatment and for some time afterwards. Some types of erythropoietin may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

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Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

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